



EXECUTIVE INSIGHTS

Elevating Medical Affairs: Seven Recommendations for High-Performing Biopharma Organizations

The commercialization environment for biopharma companies is far more complicated than it was a decade ago, and several converging forces have elevated the importance of Medical Affairs. These forces are requiring organizations to reevaluate their operating models, including engaging Medical Affairs earlier, resourcing it more deliberately, and integrating it more closely with clinical development and commercialization planning.

In this edition of L.E.K. Consulting's *Executive Insights*, we share our point of view on where high-performing Medical Affairs functions should be headed, informed by the forces reshaping biopharma commercialization and by our work helping companies build these functions. The seven recommendations that follow describe the operating model changes we expect to differentiate top performers over the next launch cycle, not the model most companies run today.

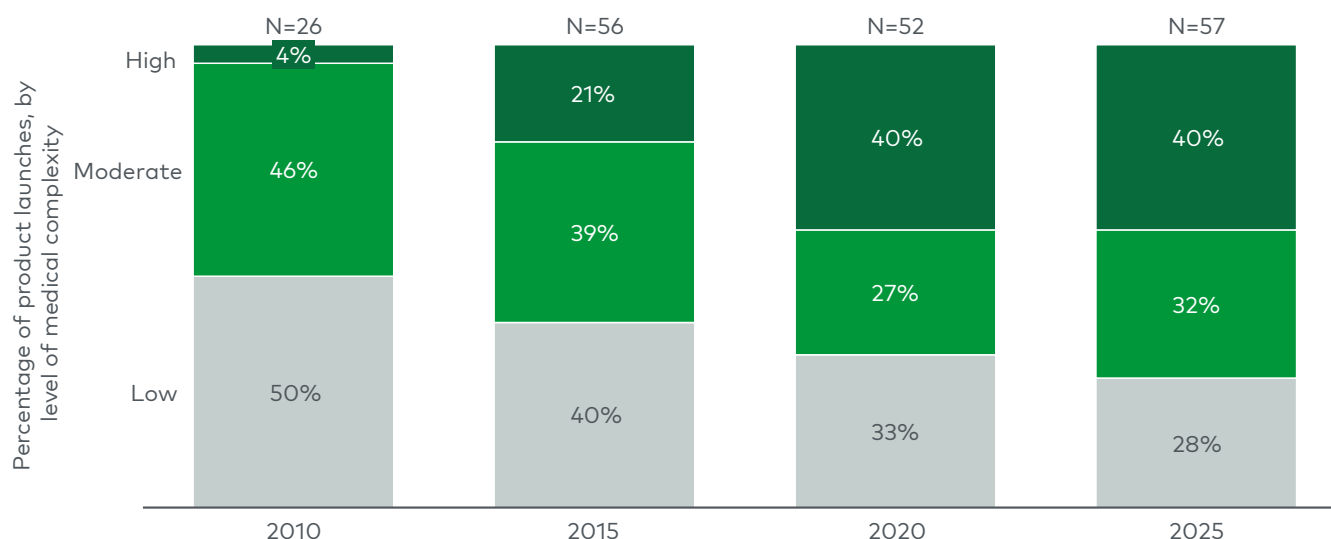
Four structural forces have made Medical Affairs a strategic priority

The launch pipeline has become more scientifically demanding. From 2010 to 2025, the share of therapies requiring sustained scientific engagement well before approval increased significantly (see Figure 1). These include novel or first-in-class products; advanced modalities like cell, gene, radioligand and bispecific therapies products with complicated care pathways; those requiring biomarker or companion diagnostics; those with challenging benefit-risk profiles (black-box warnings and off-label considerations); and those requiring extensive real-world evidence (RWE) to secure coverage and formulary access.

Figure 1

Medical Affairs importance in new product launches

Medical product launch complexity, 2010, 2015, 2020, 2025



Note: Product complexity was measured on a scale of 0 to 6 by scoring products based on whether they possessed the following traits that are indicative of complex products: novel mechanism of action, advanced modality, complex care pathway, biomarker/companion diagnostics dependency, complex benefit-risk profile, high RWE burden. Products possessing two to three traits are characterized as moderate medical complexity, while products with four or more traits are high medical complexity.

Source: L.E.K. research and analysis

The Inflation Reduction Act (IRA) has tightened the evidence timeline. The first 10 Medicare drug price negotiations produced reductions ranging from 38% to 79%,¹ with the quality of the supporting evidence package a key differentiator. Manufacturers that invest in comparative real-world data will be better positioned for future negotiation cycles than those that do not. Under the IRA, small molecules become eligible for negotiation seven years after approval. For companies with products currently in Phase II or III development, that window is much closer than it may seem. Evidence building must begin before approval, not after.

Declining Commercial representative access to physicians has increased the importance of medical science liaisons (MSLs). Access has narrowed over the past decade as integrated delivery networks have consolidated, value-based care models have expanded and product science has grown more complex. In specialty and rare disease especially, MSLs have become the primary and most trusted channel for engaging physicians.

Artificial intelligence (AI) and digital tools are widening the capability gap across organizations. Companies that have embedded AI into everyday operations such as call planning, insight synthesis, evidence analytics and content generation are already seeing productivity and quality advantages over peers still running isolated pilots.

Seven recommendations for building a high-performing Medical Affairs function

Drawing on our work with biopharma companies at different stages of commercialization, we have identified seven operating model recommendations to create high-performing Medical Affairs functions.

1. Have Medical Affairs report directly to the CEO

Most companies have a single executive, e.g., head of R&D or chief medical officer, who oversees Clinical Development, Medical Affairs and sometimes Regulatory as well. While this structure is common, it signals that Medical Affairs carries less strategic weight than Commercial or R&D, and it leaves Medical Affairs without its own direct representation at the executive level when key decisions such as pivotal trial design, launch sequencing and evidence-generation trade-offs are made.

We recommend that organizations with medically complex launches (as described above) elevate Medical Affairs to report directly and independently to the CEO, alongside R&D and Commercial, with shared impact on and accountability for pivotal trial design, go-to-market model, New Drug Application filing and evidence-generation planning and execution. This is not the norm today, but we believe it should be in biopharma companies with medically complex treatments.

2. Include Medical Affairs from Phase II

Medical Affairs is often involved too late. By the time a pivotal trial is complete, the most consequential Medical Affairs decisions, including key opinion leader (KOL) engagement strategy, advisory board composition, evidence gap assessment and payer evidence planning, are already decided or behind schedule. Leading organizations begin this work 24 to 36 months before expected approval, with Medical Affairs sharing ownership of the evidence strategy alongside Clinical Development and Pricing & Market Access.

This early involvement can take several forms. For example, some organizations bring MSLs or other medical hires in to help run clinical operations activities such as trial recruitment and to build out physician and KOL relationships, then transition them into core Medical Affairs roles as launch approaches.

To ensure proper Medical Affairs engagement, Medical Affairs should participate in development governance beginning in Phase II; Clinical Development, Medical Affairs and Commercial should jointly own the Integrated Evidence Plan and pivotal trial design; and payer-focused evidence gap assessments should start before Phase III is complete.

3. Build real-world evidence infrastructure before it is needed

In a recent survey, nearly 20% of payers say they regularly use real-world evidence (RWE) in formulary decisions and 80% would like to.² We expect RWE's weight in formulary decisions to grow over the next several years. Additionally, as manufacturers and researchers build a deeper base of comparative RWE in the Medicare population, we expect the importance of health economics and outcomes research in IRA price negotiation cycles to increase as well.

To succeed, Medical Affairs should have RWE capabilities, active data partnerships and an evidence roadmap for each pipeline asset before approval. Building this infrastructure — e.g., electronic medical record vendor partnerships, specialty pharmacy networks and disease registries — takes time to do well, which is why small companies should consider accessing it through strategic contract research organization and data vendor partnerships rather than building from scratch. Outsourcing is especially attractive where the anticipated depth and frequency of RWE engagement doesn't yet justify permanent headcount. It can put capabilities into place in months rather than the year or more required to build proprietary infrastructure in-house and preserves the option to bring the work in-house later as evidence needs scale.

4. Carefully coordinate Medical Affairs and Commercial

A lack of coordination between Medical Affairs and Commercial commonly impairs launch effectiveness. The customer experience suffers when messages land inconsistently and scientific and promotional touchpoints are poorly sequenced.

Companies should align Medical Affairs and Commercial into integrated, cross-functional teams while retaining distinct reporting lines. They should have coordinated planning, a coherent scientific and value narrative, and clear rules of engagement to allow each function to reinforce the other without compromising the independence that is critical for compliance and retaining physician trust.

5. Invest in the MSL team

With scientifically complex launches, the importance of early and significant market development increases, and fielding MSLs provides a key opportunity to build critical KOL relationships, support clinical trial enrollment, assess treatment centers, and carry out similar prelaunch activities. We recommend organizations resource the MSL team against a defined set of prelaunch milestones (e.g., KOL mapping and advisory board formation, clinical trial site engagement, and payer-facing evidence generation) rather than scaling headcount reactively as launch timing crystallizes. The right size and sequencing depend on the depth of scientific engagement a given launch requires, not a fixed ratio, industry benchmark or geographic coverage model.

Where speed matters more than continuity, for example to field a team quickly ahead of a compressed launch window, contracted MSL support can be operational faster than an in-house team. Because contracted MSLs tend to have higher turnover rates than an in-house team, we recommend using a contracted team as a bridge to ensure that the organization can establish the long-tenured relationships that enable early market education and support through launch.

6. Incorporate AI in the future-state medical affairs operating model

AI is already changing how Medical Affairs teams work, and Medical Affairs may be one of the functions where AI offers the most meaningful productivity gains. Much of the work that has historically consumed significant time for Medical Affairs staff is well suited to AI support. The clearest opportunities today are in RWE generation (e.g., cohort development, literature monitoring and report drafting) and MSL support, where AI can help teams practice scientific exchanges, find relevant literature in the field and tailor follow-ups to physician questions. AI can also support insights generation and aggregation, content adaptation for different audiences, and Medical Information request management. Adoption of AI tools can free up experienced Medical Affairs professionals for work that requires scientific judgment, clinical depth and in-person stakeholder contact. To capture these gains, Medical Affairs must define which tasks AI will handle, establish governance for AI workflows and content management, and evolve its operating model as AI adoption matures.

7. Measure Medical Affairs impact

A 2025 global survey of over 1,000 Medical Affairs professionals found that 92% of organizations still rely primarily on activity volume to assess MSL performance, even though only 3% consider their current metrics very effective.³ Instead, organizations need a measurement framework with three levels.

Activity metrics (like healthcare professional interactions, advisory boards and publications) provide operational visibility and are a reasonable starting point.

Quality metrics go a step further. They include physician feedback on the quality of scientific exchange, how often field insights influence development or market access decisions, and a determination of whether evidence gaps are closing over time against plan.

Outcome, or impact, metrics are harder to define but more meaningful. Examples include changes in physician diagnostic or treatment behavior following a Medical Affairs education program, or faster time to diagnosis in therapeutic areas where MSLs have focused on disease recognition and testing.

No metric should link MSL activity to prescription volume or sales performance. Maintaining that separation is both a compliance requirement and a prerequisite for physician trust.

What biotech executives should do now

In summary, we recommend making Medical Affairs a true third pillar alongside R&D and Commercial, built years ahead of launch to support pivotal trial design and execution, educate the market and build relationships, and establish the foundation for RWE generation. For organizations in the early stages of building or restructuring their Medical Affairs function, the decisions made in the near term may shape the function's impact across multiple launch cycles. For those with more established Medical Affairs functions, these recommendations help identify where enhancements are likely to have the greatest positive impact on the intended use of the organization's treatments.

For more information, please [contact us](#).

Endnotes

¹ISPOR.org, "HPR96 The Changing Impact of the Inflation Reduction Act on Payer Utilization Management: Insights from an Ongoing Quantitative Survey." [d3cb6533-47cf-4cd7-b37b-e930b6f67c69.pdf](https://www.ispor.org/doi/10.18553/jmcp.2025.25108)

²JMCP.org, "AMCP real-world evidence standards: Overcoming barriers to using real-world evidence in US payer decision-making." <https://www.jmcp.org/doi/10.18553/jmcp.2025.25108>

³MDPI.com, "Challenges of Key Performance Indicators and Metrics for Measuring Medical Science Liaison Performance: Insights from a Global Survey." <https://www.mdpi.com/2226-4787/13/2/51>

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